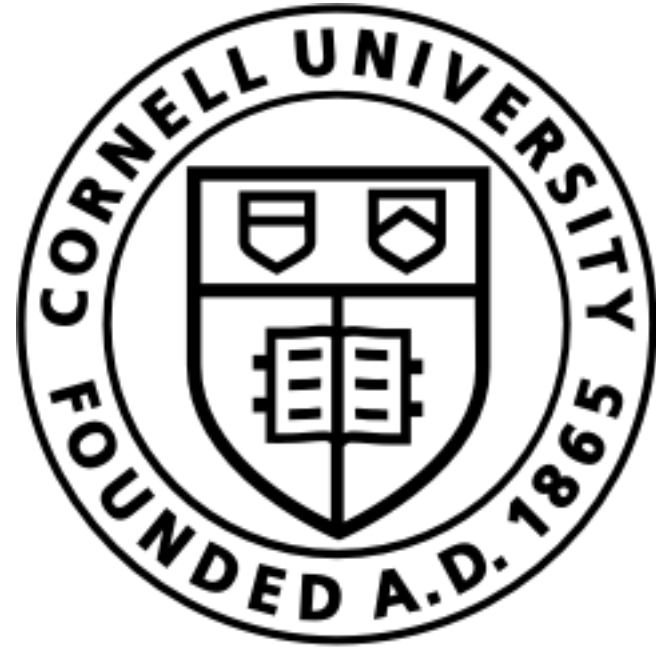




Lecture 2a: The secret meaning of marginal utility and its role in satisfaction, trading, and tradeoffs

CHEME 5760 Cornell University

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Objectives: In this lecture we will:

- ▶ **O1:** Compute the utility of a Tesla Model S versus a Honda Odyssey using the Cobb-Douglas utility function
- ▶ **O2:** Use the Marginal Utility (MU) to optimize the features of the Tesla Model S
- ▶ **O3:** Use the Marginal Utility (MU), Indifference, and the Marginal Rate of Substitution (MRS) to compute exchange opportunities

L2a notes: <https://bit.ly/44waH1E>

Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- **Example:** CHEME-5760-L2a-TeslaVersusHonda.ipynb
- **Example:** CHEME-5760-L2a-IndifferenceCurves.ipynb

Refinement: Utility Functions Redux

An individual decision-maker (agent) is given a set of $X_1, \dots, X_n$ objects where each object has m possible features $X_i = \{x_{1,i}, \dots, x_{m,i}\}$. A utility function ranks the agent's preference for combinations of features for each object i :

$$U_i(x_{1,i}, x_{2,i}, \dots, x_{n,i}) = u_i \quad (1)$$

where u_i is a real number called the utility for object i . Utility has units of utils.

- The utility function $U : X \rightarrow \mathbb{R}$ is unique only up to an order-preserving transformation.
- Utility functions are ordinal, i.e., they rank-order bundles but do not measure differences between bundles.

Tesla model S

Link: <https://shop.tesla.com/>



Honda Odyssey

Link: <https://automobiles.honda.com/odyssey>

Scoring system



Features	Exponents	Tesla	Honda
Sustainability	0.2	5	3
Affordability	0.1	2	4
Styling	0.05	5	3
Usefulness	0.3	2	5
Cost of Ownership	0.1	4	2
Performance	0.05	5	1
Safety	0.2	5	5

O1: Compute the utility of a Tesla Model S versus a Honda Odyssey using the Cobb-Douglas utility function

Utility Function: The Cobb-Douglas Utility

The Cobb-Douglas utility function is the product of the m feature variables, thus, it models situations where features occur simultaneously. Each feature variable is raised to a non-negative exponent:

$$U(x_1, \dots, x_m) = \prod_{i \in 1 \dots m} x_i^{\alpha_i} \quad (3)$$

In our realization of the Cobb-Douglas utility, the exponents must sum to unity $\sum_{i \in 1 \dots m} \alpha_i = 1$, $x_i \geq 0$, and $\alpha_i \geq 0$.

[switch to the Honda/Tesla notebook example]

O2: Use the Marginal Utility (MU) to optimize the features of the Tesla Model S

Concept: Marginal Utility

Let x_i (for $i \in 1 \dots n$) be a feature, e.g., a good, service or a state of the world which is a member of the set X . Further, let $U : X \rightarrow \mathbb{R}$ be a utility function over the set X . The **marginal utility** of $x \in X$ is the partial derivative of U with respect to x :

$$\bar{U}_{x_i} \equiv \left(\frac{\partial U}{\partial x_i} \right) \quad (2)$$

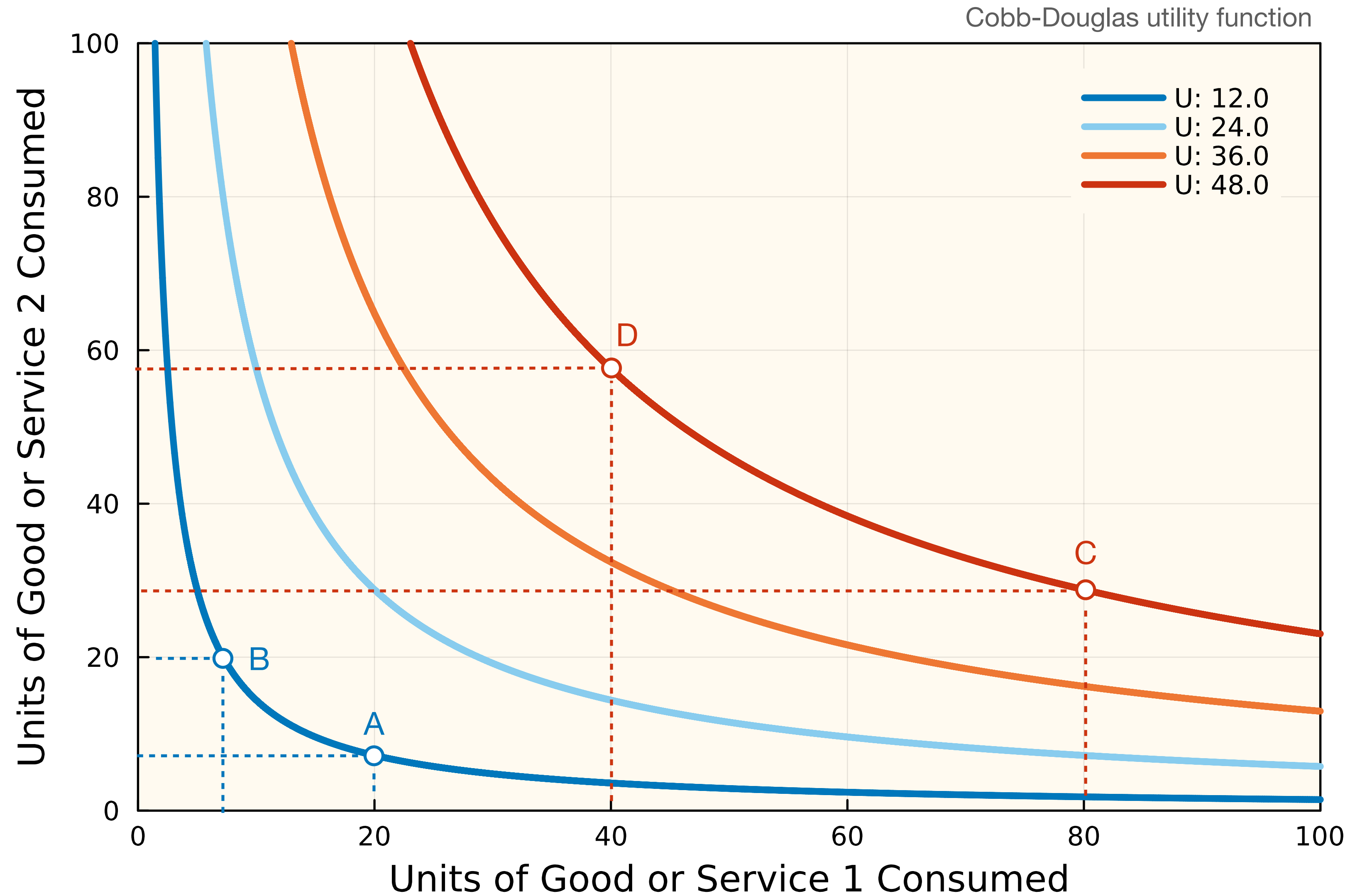
The marginal utility $\bar{U}_{x_i}$ measures the change in the utility U resulting from small changes in the feature variables x_i .

Marginal utility in words: The additional satisfaction gained by a small increase in a feature variable, e.g., the increase in utility resulting from a +1 unit increase in the fuel efficiency of your car

[switch to the Honda/Tesla notebook example]

O3: Use the Marginal Utility (MU), Indifference, and the Marginal Rate of Substitution (MRS) to compute exchange opportunities

Indifference curves. An indifference curve is a curve of *constant utility* in the space of objects or options the decision-maker chooses. Any two points on an indifference curve have the same utility.



[switch to the Indifference Notebook example]

Concept: Local Utility Expansion

An individual decision-maker (agent) is given a set of $X_1, \dots, X_n$ objects where each object has m possible features $X_i = \{x_{1,i}, \dots, x_{m,i}\}$. The *local change* in the utility $U(\dots)$ near a point $x^* \in X$ can be computed using the *total differential* of the utility function $U(\dots)$:

$$dU = \sum_{i \in 1 \dots m} \bar{U}_{x_i} \cdot dx_i \quad (4)$$

where $dU \approx (U - U^*)$ denotes the local change in utility, $\bar{U}_{x_i}$ denotes the marginal utility of x_i evaluated at the point $x^* \in X$, and $dx_i \approx (x_i - x_i^*)$ is the change in the feature x_i .

[switch to the Honda/Tesla notebook example]

Concept: Marginal Rate of Substitution and Indifference

The marginal rate of substitution describes the rate at which a consumer is willing to trade one good (or feature) for another. For a utility function $U : X \rightarrow \mathbb{R}$, the marginal rate of substitution between any two goods (or features) x_i and x_j along an *indifference curve* with utility U^* , i.e., a curve with utility U^* and $dU = 0$, is given by:

$$\bar{U}_{x_i} dx_i = -\bar{U}_{x_j} dx_j \quad \forall i \neq j \quad (5)$$

or equivalently:

$$\frac{dx_i}{dx_j} = -\frac{\bar{U}_{x_j}}{\bar{U}_{x_i}} \quad \forall i \neq j \quad (6)$$

[switch to the Honda/Tesla notebook example]

Today?